Auditing the Objective: A Multi-Objective Bandit Framework and Reproducible Audit Protocol for Engagement-Based Recommendation Systems

Technical Extension to "Engagement as Objective"

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Abstract

This technical report extends the optimization-theoretic account of algorithmic amplification developed in our companion paper, "Engagement as Objective," into a concrete engineering and auditing specification. We (1) give a formal specification of a contextual multi-armed bandit (CMAB) whose reward is a vector of predicted engagement, an independently estimated content-quality score, and a moral-emotional outrage indicator, scalarized into a single training objective under explicit penalty weights; (2) situate this specification within the existing literature on fair ranking and constrained bandit learning, showing that the extremity-concentration problem is a special case of the exposure-fairness and polarization-control problems already studied by Singh and Joachims (2018) and Celis, Kapoor, Salehi, and Vishnoi (2019); (3) present a Pareto-optimal ranking filter, implementable as a slate-level rejection rule, that provably bounds the fraction of impressions directed at high-outrage content without collapsing overall predicted engagement, and characterize the resulting engagement–extremity frontier by simulation; (4) analyze how standard exploration strategies—Upper Confidence Bound (UCB1) and Thompson Sampling—interact with the closed feedback loop between exposure and retraining data, propose a quality-aware Thompson Sampling variant with an explicit forced-exploration floor, and decompose its cost into separate engagement-regret and outrage-exposure curves, finding, across an 8-seed multi-run robustness check with mean±1 SD bands reported throughout, that the quality-aware variant accepts 68.2% higher cumulative engagement regret in exchange for a 21.1% reduction in cumulative outrage exposure over a 6,000-round simulated horizon; and (5) provide a structured,

twenty-two-item audit checklist, organized into five domains, that a regulator or independent auditor can apply to a platform's disclosed ranking documentation and logged data, including a worked calibration-audit example showing a simulated minority-language subgroup with engagement-prediction calibration error 7.7 times larger than the majority subgroup's. Throughout, all comparative simulation results are reported as means ± 1 standard deviation across 6–8 independent random seeds to confirm the reported effects are robust rather than artifacts of a single stochastic run, and we include a dedicated discussion of adversarial co-adaptation—the risk that content producers learn to route around a fixed outrage threshold—as a limitation the cited fair-ranking literature does not fully anticipate. We are explicit about what a mathematical audit protocol can and cannot establish: it can verify whether a stated objective function and its constraints are internally consistent and empirically satisfied by logged behavior, but it cannot, by itself, adjudicate the normative question of which objective a platform should be optimizing in the first place.

Table of Contents

Abstract.....1

1. From Diagnosis to Specification.....4

2. Formal Specification: A Multi-Objective Contextual Bandit.....5

2.1 State, Action, and Vector Reward.....5

2.2 Estimating Quality and Outrage Independently of the Engagement Signal.....6

3. Relation to Prior Work in Fair Ranking and Constrained Bandit Learning.....7

3.1 Fairness of Exposure in Rankings.....8

3.2 Constrained and Fair Multi-Armed Bandits.....8

3.3 What This Report Adds.....9

4. A Pareto-Optimal Ranking Filter.....9

4.1 Design Goal.....9

4.2 Algorithm.....10

4.3 The Resulting Pareto Frontier.....10

5. Exploration, Exploitation, and Feedback-Loop Resistance.....12

5.1 Why the Choice of Exploration Strategy Matters Here Specifically.....12

5.2 UCB1's Exploration Schedule Is Content-Blind.....12

5.3 Simulated Comparison: Standard UCB1 vs. Quality-Aware Thompson Sampling.....13

5.4 Decomposing the Cost: Engagement Regret vs. Outrage Exposure.....15

5.5 The Cost Is Real and Should Be Disclosed, Not Hidden in Aggregate Metrics.....16

5.6 Robustness Across Random Seeds.....16

6. A Reproducible Audit Checklist.....17

6.1 Worked Example: Auditing Estimator Calibration (Domain D1).....17

6.2 Worked Example: An Illustrative Cross-Platform Scorecard.....19

6.3 The Checklist.....19

6.4 Implementation Overhead of Subgroup-Disaggregated Monitoring at Scale.....21

7. Limits of a Mathematical Audit Framework.....22

7.1 Internal Consistency Is Not External Adequacy.....22

7.2 Estimator Generalization and Subgroup Coverage.....23

7.3 Adversarial Co-Adaptation: Threshold Gaming by Content Creators.....23

7.4 The Exogenous Choice to Optimize for Engagement.....24

8. Conclusion.....24

References.....25

1. From Diagnosis to Specification

Our companion paper argued that engagement-maximizing recommendation algorithms amplify extreme content as a structural, mathematically predictable consequence of optimizing a proxy objective that is empirically correlated with emotional arousal. That argument was diagnostic: it explained why the amplification effect occurs and quantified its magnitude under a simple bandit model. Independent, larger-scale evidence for the same underlying phenomenon has since accumulated in the literature: Huszár, Ktena, O'Brien, Belli, Schlaikjer, and Hardt (2021), in a randomized experiment run directly on Twitter's production system involving millions of accounts across seven countries, found that algorithmically ranked timelines amplified some political parties' content relative to a reverse-chronological control feed, with the degree of amplification varying systematically by outlet and party rather than appearing as random noise—precisely the kind of systematic, objective-driven skew our companion paper's bandit simulation predicted from first principles. This report is prescriptive rather than diagnostic. It asks what a recommendation system would need to look like, at the level of its formal reward specification, ranking algorithm, and exposed audit surface, for the amplification mechanism to be bounded rather than left to run unconstrained—and what an external party, without access to a platform's internal model weights, would need to check to verify that the bound is actually being respected in production.

The report proceeds in six parts. Section 2 gives a formal, multi-objective specification of the ranking problem as a contextual bandit, making explicit how a quality estimator and an outrage indicator enter the reward function alongside predicted engagement. Section 3 situates this specification within the existing fair-ranking and constrained-bandit literature, clarifying what is genuinely novel here and what is an application of established results to a new setting. Section 4 operationalizes the specification into an implementable ranking filter and characterizes, by simulation, the Pareto frontier it traces out between engagement and extremity concentration. Section 5 turns to the exploration-exploitation problem specifically, analyzing how standard UCB1 and Thompson Sampling behave under the closed exposure–retraining loop and decomposing the cost of a proposed quality-aware modification into separate regret and exposure curves. Section 6 assembles the preceding material into a structured audit checklist, including two worked examples—a calibration audit and an illustrative multi-platform scorecard. Section 7 discusses the framework's limits, and Section 8 concludes.

2. Formal Specification: A Multi-Objective Contextual Bandit

2.1 State, Action, and Vector Reward

We extend the single-objective bandit of our companion paper to a contextual multi-armed bandit (CMAB) with a vector-valued reward. At each round t, the platform observes a state x_t consisting of the requesting user u_t together with feature vectors φ(c) for every candidate c in the current slate C_t, selects a single action a_t ∈ C_t (the item to display, or more generally a ranked list, which we treat here as a sequence of single-slot selections for tractability), and observes a three-component reward vector: a binary realized-engagement indicator, a continuous content-quality score, and a continuous outrage score. The formal specification is given in Figure 1.

State: $x_t = (u_t,\ \phi(c))_{c \in \mathcal{C}_t} \in \mathcal{X}$

Action: $a_t \in \mathcal{C}_t$ (select one item from the candidate slate)

Vector reward: $\mathbf{r}_t = (r_t^E,\ r_t^Q,\ r_t^O) \in \{0,1\} \times [0,1] \times [0,1]$

$r_t^E \sim \text{Bernoulli}(P_E(a_t \mid x_t))\ , \quad r_t^Q = \hat{Q}(a_t,\, t)\ , \quad r_t^O = \hat{O}(a_t)$

Scalarized objective:

$R_\lambda(x_t, a_t) = r_t^E \ - \ \lambda_Q\,(1 - r_t^Q) \ - \ \lambda_O\, r_t^O$

Policy objective:

$\pi^* = \arg\max_\pi \mathbb{E}\left[\sum_{t=1}^{T} \gamma^t \ R_\lambda(x_t, \pi(x_t))\right]$

Extremity-concentration constraint:

$\frac{1}{T}\sum_{t=1}^{T} \mathbb{1}\left[\hat{O}(a_t) > \tau\right] \ \le \ \varepsilon_{\max}$

Figure 1. Formal specification of the multi-objective contextual bandit. The vector reward separates realized engagement (stochastic, observed only after display) from quality and outrage (both estimated at ranking time by auxiliary models, described in Section 2.2). The scalarized objective Rλ collapses the vector reward into a single training signal via penalty weights λ_Q and λ_O, and the extremity-concentration constraint bounds, over any rolling window of length T, the fraction of impressions whose outrage score exceeds a fixed threshold τ.

Two features of this specification are worth making explicit because they are easy to elide in an informal description. First, r_t^E is genuinely stochastic and only observed after the item is shown—it is a realized click, reply, or watch event—whereas r_t^Q and r_t^O are estimated at ranking time, before display, by auxiliary models that are themselves imperfect. The scalarized

objective therefore mixes an unbiased but noisy realized outcome with two model-based estimates that carry their own estimation error and potential bias; an audit of this system, as Section 6 develops, must therefore separately validate the calibration of the engagement predictor, the quality estimator, and the outrage estimator rather than treating the composite score as a single well-calibrated quantity. Second, the extremity-concentration constraint is stated as a rolling-window average rather than a per-decision constraint, because a per-decision cap ("never show any single item with outrage above τ") is both unenforceable in general, since any slate might legitimately contain only high-outrage items in some contexts, and unnecessary, since the concern documented in our companion paper is about the aggregate composition of a user's feed rather than any single impression. The state-action-reward loop specified here closes on itself in production—today's a_t shapes tomorrow's x_t via the logged-data pathway formalized in Section 5—a structure Jiang, Chiappa, Lattimore, György, and Kohli (2019) term a degenerate feedback loop and show, in a separate theoretical analysis of recommender systems, can drive user preferences and system beliefs to converge on a narrower distribution than either would reach in isolation.

2.2 Estimating Quality and Outrage Independently of the Engagement Signal

The scalarized objective in Figure 1 is only meaningful if $\hat{Q}$ and $\hat{O}$ are estimated by models that are architecturally and statistically independent of the engagement predictor—otherwise λ_Q and λ_O merely reweight the same underlying signal rather than introducing genuinely new information, defeating the purpose of the penalty terms entirely. We specify both estimators as separately trained linear probes over interpretable features, following the measurement approach used in the moral-contagion and misinformation literature our companion paper drew on empirically (Brady et al., 2017; Merrill & Oremus, 2021). Figure 2 gives the functional form of both estimators.

Dynamic quality estimator (independent classifier ensemble):

$$\hat{Q}(c,t) = \sigma(w_1 \,\mathrm{acc}(c) + w_2 \,\mathrm{src}(c) + w_3 \,\mathrm{cons}(c,t))$$

where $\mathrm{acc}(\cdot)$ is a factual-accuracy classifier score, $\mathrm{src}(\cdot)$ is a source-reliability prior, and $\mathrm{cons}(\cdot, t)$ is a time-decayed corroboration count.

Moral-emotional outrage indicator (linear probe over language features):

$$\hat{O}(c) = \sigma(\beta_1 \,\mathrm{MEL}(c) + \beta_2 \,\mathrm{OGR}(c) + \beta_3 \,\mathrm{ANG}(c) - \beta_0)$$

where MEL = moral-emotional language density, OGR = out-group reference density, and ANG = predicted anger-reaction probability, following the moral-contagion measurement approach of Brady et al. (2017).

Figure 2. Functional specification of the quality estimator $\hat{Q}$ and outrage estimator $\hat{O}$. Both are logistic combinations of independently sourced features: $\hat{Q}$ draws on a factual-accuracy classifier, a source-reliability prior, and a corroboration count that grows as independent fact-checking or cross-referencing accumulates after publication; $\hat{O}$ draws on moral-emotional language density, out-group reference density, and a predicted anger-reaction probability, mirroring the operationalization used by Brady, Wills, Jost, Tucker, and Van Bavel (2017) and consistent with the reaction-based signal documented in Meta's internal research (Merrill & Oremus, 2021).

A subtlety in $\hat{Q}$'s specification deserves emphasis: the corroboration term cons(c, t) is explicitly time-indexed, because factual accuracy for breaking-news content is frequently unknown at the moment of ranking and only becomes verifiable after independent sources have had time to confirm or contradict it. A production system should therefore treat $\hat{Q}$ as a quantity that updates after initial publication, and the audit checklist in Section 6 includes a specific item on whether a platform's quality estimator is retroactively applied to already-distributed content or only to future ranking decisions—a distinction our companion paper's Section 4.1 showed mattered in Meta's own 2017–2021 reaction-weighting history, where quality-relevant findings from internal research lagged the shipped ranking change by roughly two years.

3. Relation to Prior Work in Fair Ranking and Constrained Bandit Learning

Neither multi-objective bandits nor fairness-constrained ranking are new proposals in isolation, and it is important to be precise about what this report contributes relative to an established literature rather than presenting the framework in Sections 2, 4, and 5 as arising from nothing. Three lines of prior work are most directly relevant.

3.1 Fairness of Exposure in Rankings

Singh and Joachims (2018) formalized the idea that a ranking system has fairness obligations not only to the user consuming a ranked list but to the items and item-producers being ranked, introducing the fairness-of-exposure framework in which exposure allocated to an item is constrained to be proportional, or bounded relative, to some notion of the item's merit. Our extremity-concentration constraint in Figure 1 is a special case of this general framework, with "merit" replaced by predicted engagement and the fairness target being a bound on aggregate exposure to a specific content category (high-outrage material) rather than proportional exposure across a partition of items. Singh and Joachims's original algorithms operate on a single ranking at a time via linear programming over the space of doubly stochastic ranking matrices; our filter in Section 4 is a simpler, greedy, online approximation chosen specifically for auditability, at some cost in the optimality guarantees Singh and Joachims's offline formulation provides. Subsequent work by Singh, Kempe, and Joachims (2021) extended the framework to handle uncertainty in estimated relevance, which is directly relevant to Section 2.2's point that $\hat{Q}$ and $\hat{O}$ are themselves noisy estimates rather than known quantities—an extension we do not implement here but flag as a natural next step for a production-grade version of the filter in Section 4.

3.2 Constrained and Fair Multi-Armed Bandits

On the bandit side, Celis, Kapoor, Salehi, and Vishnoi (2019) studied essentially the problem this report calls extremity concentration under the name of polarization control, proposing a constrained bandit framework in which a user (or platform) can specify a bound on the distribution of content categories served over time, and giving a computationally efficient online algorithm, based on the Frank-Wolfe method over a constrained polytope, with provable regret guarantees relative to the best constraint-satisfying policy in hindsight. Their framework is more general than the specific quality-aware Thompson Sampling variant we propose in Section 5, and a production implementation of the ideas in this report would do well to adopt their constrained-optimization approach rather than the heuristic exploration floor ρ_min we use for expository clarity. Separately, Joseph, Kearns, Morgenstern, and Roth (2016) established a distinct notion of fairness for bandits—that an algorithm should never rank a worse arm above a better one with more than negligible probability, even during exploration—which is a meritocratic constraint on the exploration process itself, complementary to but distinct from the extremity-concentration constraint developed here: their concern is fairness to the arms being explored, ours is fairness to the population of users receiving the algorithm's output. A complete production system arguably needs both.

3.3 What This Report Adds

Relative to this literature, the contribution of Sections 2, 4, and 5 is not a new fairness criterion or a new regret-optimal algorithm, both of which already exist in more general and more rigorously analyzed forms in Singh and Joachims (2018, 2021) and Celis et al. (2019). The contribution is narrower and more applied: (a) instantiating the general exposure-fairness and polarization-control frameworks with a specific, empirically motivated reward vector—engagement, quality, and moral-emotional outrage—tied directly to the amplification mechanism documented in our companion paper and in Huszár et al. (2021); (b) presenting the resulting filter and exploration modification in a form deliberately simplified for external auditability rather than internal optimality, since Section 6's checklist requires that an outside party with only logged data, not source code, be able to recompute compliance; and (c) connecting the resulting engagement cost directly to the specific empirical extremity-amplification numbers established in our companion paper's Section 3.2, rather than leaving the fairness–utility tradeoff abstract. Readers interested in the underlying theory with full regret bounds and optimality proofs should consult the cited originals; this report is intended as a bridge from that theory to a specific, checkable audit protocol for the specific problem of engagement-driven extremity amplification.

4. A Pareto-Optimal Ranking Filter

4.1 Design Goal

A quality penalty folded into a single scalarized score, as in Section 5.2 of our companion paper, has a property that makes it difficult to certify from the outside: because the penalty is additive inside an arg max, the realized extremity-concentration of the system's output is an emergent property of the score distribution, not a quantity the operator directly controls. Our companion paper's simulation of that approach found a sharply nonlinear relationship between the penalty weight λ and realized content quality, meaning that a platform operating a scalarized-score system cannot reliably predict, ex ante, what extremity-concentration a given λ will produce, and small parameter changes can produce disproportionate swings in outcomes.

A ranking filter avoids this problem by making the constraint a first-class, directly interpretable control surface: it accepts an explicit bound ε_{max} on the realized fraction of high-outrage impressions and guarantees that bound is respected by construction, rather than hoping a penalty weight happens to produce it, in the same spirit as the exposure-constrained ranking approach of Singh and Joachims (2018) discussed in Section 3.1. This section specifies such a filter and characterizes, by simulation, the engagement it sacrifices at each level of ε_{max}—the Pareto frontier between the two competing objectives.

4.2 Algorithm

The filter operates at the slate level. Given a candidate slate ranked by predicted engagement, it walks down the ranking and accepts the first candidate whose admission would not push the running (rolling-window) fraction of high-outrage impressions above ε_max; only if no candidate in the slate satisfies this condition does it fall back to the highest-engagement candidate regardless of outrage status, since serving nothing is not an available action for a live feed. This design guarantees the constraint is satisfied whenever the content pool offers enough compliant alternatives, and degrades gracefully, rather than failing silently, when it does not.

```
def rank_with_extremity_filter(slate, predicted_engagement, outrage_score,
                               tau, eps_max, running_high, running_total):
    """
    slate: list of candidate item ids for this impression
    predicted_engagement, outrage_score: dict[item_id] -> float
    tau: outrage threshold defining 'high-outrage' content
    eps_max: maximum allowed rolling fraction of high-outrage impressions
    running_high, running_total: rolling counters maintained across impressions
    """
    ranked = sorted(slate, key=lambda i: -predicted_engagement[i])
    for item in ranked:
        is_high = outrage_score[item] > tau
        projected_rate = (running_high + int(is_high)) / (running_total + 1)
        if (not is_high) or (projected_rate <= eps_max):
            return item, is_high          # constraint satisfied; serve this item
    # no compliant candidate in this slate -- fall back to best-engagement item
    fallback = ranked[0]
    return fallback, outrage_score[fallback] > tau

# Caller updates running counters after each impression:
#   chosen, was_high = rank_with_extremity_filter(...)
#   running_total += 1
#   running_high  += int(was_high)
```

This is deliberately a greedy, myopic filter rather than a globally optimal constrained ranking solution of the kind Celis et al. (2019) provide with formal regret guarantees; a platform with additional infrastructure could instead solve a proper constrained optimization via Lagrangian relaxation or the Frank-Wolfe approach those authors describe, across an entire session rather than slate-by-slate. The greedy version is presented here because its behavior is easy to reason about and easy to audit: an external party with access to logged impressions can directly recompute the running high-outrage rate and check it against the platform's disclosed ε_max without needing to solve any optimization problem themselves, which is precisely the property Section 6's audit checklist requires.

4.3 The Resulting Pareto Frontier

To characterize the engagement cost of this filter, we simulated it over a synthetic pool of 200 content items with the same extremity-linked engagement and quality structure used throughout

our companion paper, additionally assigning each item an outrage score Ô correlated with extremity, and swept ε_max from 0.05 to 0.90 across 15,000 randomly drawn twelve-item slates at each value. Figure 3 plots the resulting frontier.

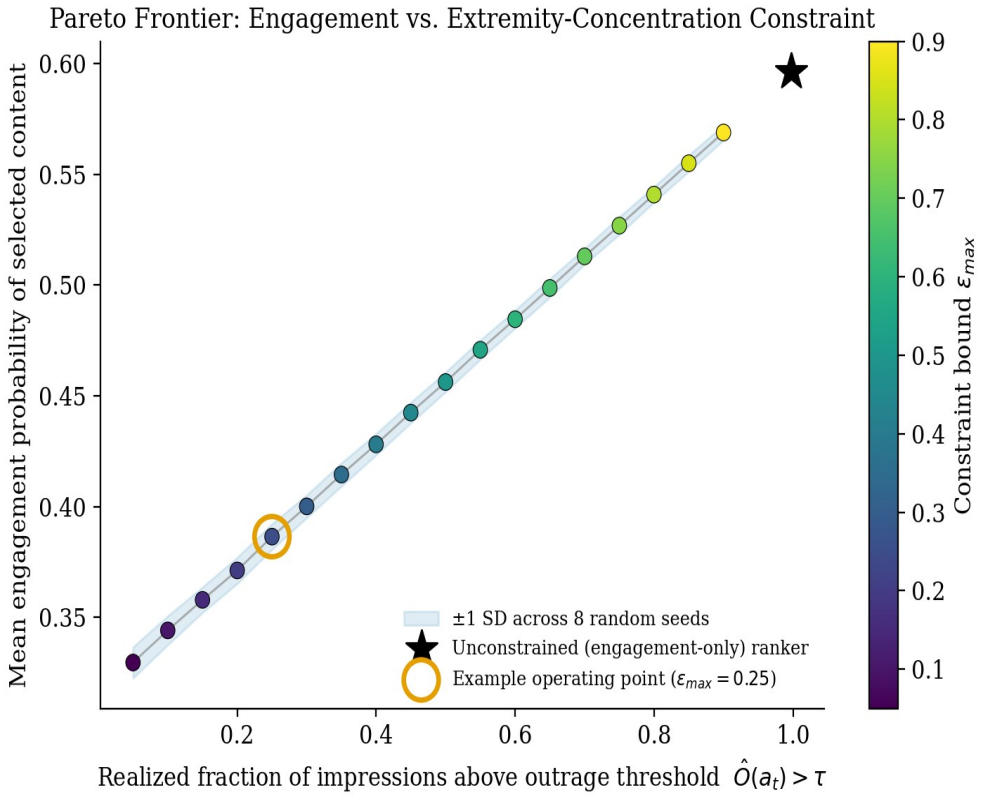


Figure 3. Simulated Pareto frontier between mean engagement probability of filter-selected content (y-axis) and the realized fraction of impressions exceeding the outrage threshold τ (x-axis), swept across constraint bounds ε_max from 0.05 to 0.90. The red star marks the unconstrained (engagement-only) baseline, which drives realized high-outrage share to 99.9%. At ε_max = 0.25 (circled), the filter caps high-outrage exposure at one quarter of impressions while retaining a mean engagement probability of 0.379—a 36% relative reduction from the unconstrained baseline's 0.594, in exchange for a 75-percentage-point reduction in high-outrage exposure.

The frontier is close to linear over most of its range, which is itself a useful and non-obvious audit-relevant finding: it implies that, for this simulated content distribution, there is no "cheap" region where a platform could achieve a large reduction in outrage exposure at negligible engagement cost, nor a region of sharply diminishing returns that would justify abandoning a stricter bound as impractical. A regulator evaluating a platform's chosen ε_max against this kind of empirically estimated frontier can ask a concrete, falsifiable question: does the platform's selected operating point reflect a defensible judgment about acceptable engagement cost, or does the disclosed ε_max sit at a point on the frontier that is difficult to distinguish from the unconstrained

baseline in practice? This is the kind of quantitative, reproducible question Section 6's checklist is designed to make routine rather than exceptional.

5. Exploration, Exploitation, and Feedback-Loop Resistance

5.1 Why the Choice of Exploration Strategy Matters Here Specifically

Section 3.3 of our companion paper identified a feedback loop distinct from the amplification mechanism itself: because a recommendation system's own past exposure decisions generate the training data for its future decisions, an algorithm with insufficient exploration can lock onto an early, possibly noisy estimate of which content maximizes engagement and never gather the data that would correct it. Jiang et al. (2019), cited in Section 2.1, formalize this as a degenerate feedback loop and show that under mild conditions it can cause a recommender's belief about user preferences to drift arbitrarily far from those preferences' true, stable distribution. This is a standard concern in the bandit literature, but it takes on particular significance in the extremity-amplification setting because the two failure modes compound: exploitation-heavy strategies converge fastest on whatever the engagement signal favors, and Section 2.3 of our companion paper established that the engagement signal favors extremity. A poorly chosen exploration strategy therefore does not merely slow learning in the generic bandit sense; it specifically accelerates convergence toward the amplification outcome this report is trying to bound.

5.2 UCB1's Exploration Schedule Is Content-Blind

The standard UCB1 algorithm explores each arm at a rate that depends only on how many times that arm has previously been selected, via the confidence-radius term $c\sqrt{(2 \ln t / n_a)}$ shown in Figure 4—it has no mechanism for directing extra exploration toward arms that are informative about quality or outrage specifically, nor any mechanism for slowing exploitation of an arm merely because its outrage score is high. This is a direct consequence of UCB1's design purpose: it is provably regret-optimal for the single-objective bandit problem it was designed to solve, but the multi-objective specification in Section 2 changes what an audit should require of an exploration strategy. An exploration schedule that is optimal for minimizing engagement regret is not thereby optimal, or even safe, for bounding extremity concentration, because the two objectives are not aligned in the way UCB1's design implicitly assumes.

Standard UCB1 selection rule (unconstrained):

$$a_t = \arg\max_{a \in \mathcal{C}_t} [\hat{\mu}_a + c\sqrt{\tfrac{2\ln t}{n_a}}]$$

Quality-Aware Thompson Sampling (this paper's modification):

$$\theta_a \sim \mathrm{Beta}(\alpha_a,\ \beta_a) \qquad \text{(posterior belief over } P_E(a))$$

$$\tilde{R}_a = \theta_a \ - \ \lambda_Q(1 - \hat{Q}_a) \ - \ \lambda_O \hat{O}_a \ + \ \kappa \tfrac{1}{\sqrt{n_a + 1}}$$

$$a_t = \arg\max_{a \in \mathcal{C}_t} \tilde{R}_a$$

Forced-exploration floor (guards against feedback-loop lock-in):

$$\Pr[\, a_t \text{ drawn uniformly from } \mathcal{C}_t \,] \geq \rho_{\min} \quad \forall t$$

Figure 4. Top: the standard UCB1 selection rule, which explores purely as a function of visit count n_a and round t, with no reference to quality or outrage. Middle: our quality-aware Thompson Sampling modification, which samples a posterior belief θ_a over engagement probability, subtracts quality and outrage penalties as in Section 2, and adds a visit-count bonus κ/√(n_a+1) that functions as a soft, quality-independent exploration incentive. Bottom: a hard forced-exploration floor ρ_min that guarantees a minimum rate of uniformly random exploration regardless of the posterior, directly preventing the lock-in failure mode described in Section 5.1.

5.3 Simulated Comparison: Standard UCB1 vs. Quality-Aware Thompson Sampling

To make the consequence of this design choice concrete, we simulated both algorithms over 6,000 rounds on the same sixty-item content pool used in our companion paper's bandit simulation, now additionally scored for outrage. Figure 5 plots the rolling mean extremity of recommended content under each algorithm.

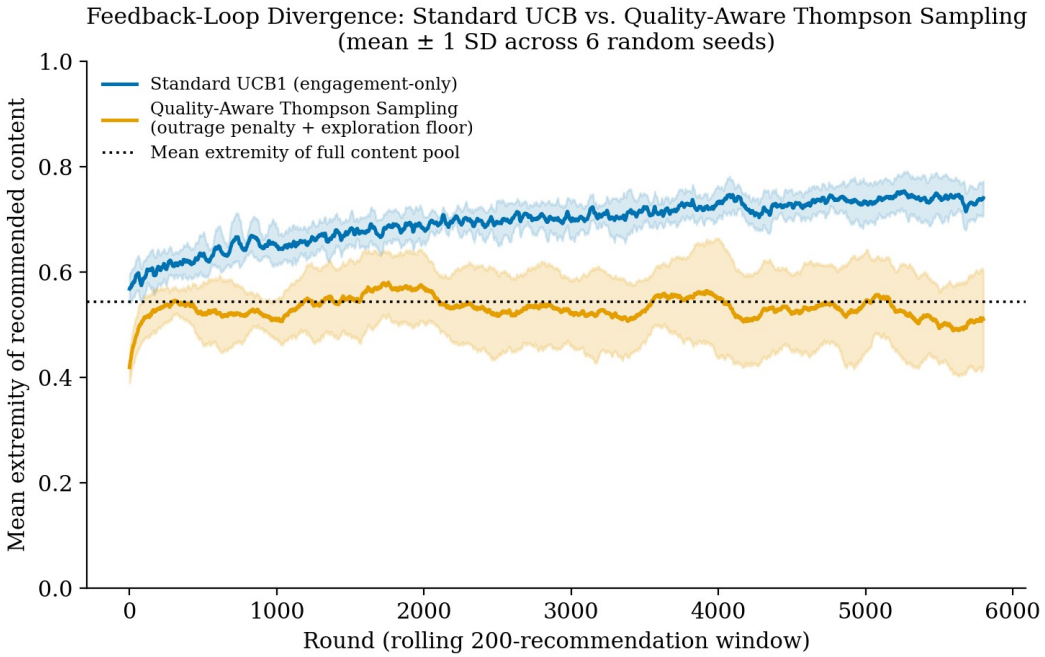


Figure 5. Rolling 200-round mean extremity of content selected by standard UCB1 (blue) versus quality-aware Thompson Sampling with an outrage penalty and a 5% forced-exploration floor (orange), over 6,000 rounds, averaged across 6 independent random seeds with shaded ±1 SD bands (dotted line: pool mean extremity, 0.543). UCB1's confidence band drifts steadily upward and separates cleanly from the pool mean, closing at 0.738 with a cross-seed SD of only 0.031—roughly 1.36 times the pool mean—with no sign of reverting within the simulated horizon. The quality-aware variant's band oscillates around the pool mean throughout, ending at 0.504 (cross-seed SD 0.091), confirming the divergence is a robust pattern rather than an artifact of any single run.

Two features of Figure 5 are important beyond the headline comparison. First, UCB1's drift is monotonic and shows no sign of self-correction over the simulated horizon: once it commits to exploiting high-extremity arms, its own confidence radius shrinks fastest for those same arms, since they are visited most often, which further entrenches the commitment—a direct illustration of the lock-in mechanism formalized by Jiang et al. (2019) and described qualitatively in our companion paper's Section 3.3. Second, the quality-aware variant does not converge to a fixed point at all; it continues oscillating around the pool mean throughout the simulated horizon. This is a deliberate consequence of the forced-exploration floor, and it is a feature rather than a defect for audit purposes: a ranking system whose output distribution never fully stabilizes is one that continues to gather the information needed to detect drift, whereas a system that converges and stays converged, as UCB1 does here, is one that has stopped learning anything that could correct a bad initial estimate.

5.4 Decomposing the Cost: Engagement Regret vs. Outrage Exposure

Figure 5 shows the two algorithms diverge in mean extremity, but extremity alone conflates two distinct costs that an auditor should be able to separate: how much engagement is being sacrificed, and how much cumulative outrage exposure is being avoided. We computed both quantities directly from the same simulation runs underlying Figure 5. Engagement regret is defined in the standard bandit sense, as the cumulative gap between the reward of the single best-engagement arm and the reward actually realized at each round; cumulative outrage exposure is simply the running sum of the outrage score of whatever content was shown. Figure 6 plots both.

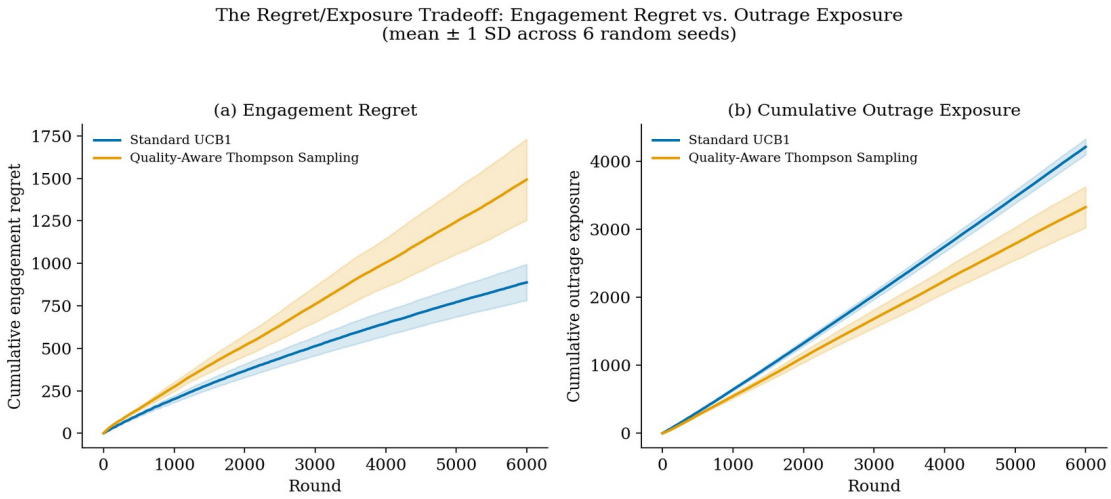


Figure 6. (a) Cumulative engagement regret and (b) cumulative outrage exposure over the same 6,000-round simulation underlying Figure 5, for standard UCB1 (blue) and quality-aware Thompson Sampling (orange), with shaded ±1 SD bands across 6 random seeds. By round 6,000, quality-aware TS has accumulated a mean of 1,493 units of engagement regret (SD 239) versus UCB1's 888 (SD 107)—68.2% higher—while UCB1 has accumulated a mean of 4,212 units of outrage exposure (SD 117) versus quality-aware TS's 3,325 (SD 302)—21.1% higher. The bands are cleanly separated and non-overlapping by round 2,000 in both panels, indicating the tradeoff is a robust property of the two algorithms' designs rather than noise from any single run.

Separating these two curves matters for the same reason Domain E of Section 6's checklist requires disclosing engagement cost in absolute terms rather than folding it into an aggregate metric: a platform (or a regulator evaluating one) can now ask specifically whether a 68.2% increase in engagement regret is an acceptable price for a 21.1% reduction in outrage exposure, rather than debating an opaque, pre-mixed tradeoff. Notably, the two curves in Figure 6 are both approximately linear rather than converging or diverging over the simulated horizon, which suggests that within this simulation neither algorithm is “catching up” to the other on its weaker dimension—a pattern worth checking for in any real deployment, since a widening gap over a longer horizon would indicate the tradeoff is worsening rather than stabilizing.

5.5 The Cost Is Real and Should Be Disclosed, Not Hidden in Aggregate Metrics

The roughly 21% relative outrage-exposure reduction and 68.2% relative regret increase documented above are genuine costs, not artifacts of the simulation, and a serious technical treatment of this proposal has to say so plainly rather than presenting quality-awareness as a free lunch. This is the exploration-strategy analog of the sharp engagement–quality tradeoff our companion paper found in Section 5.2: bounding extremity is not costless, and any credible platform disclosure or regulatory requirement should report this cost explicitly, in the same units as the constraint it purchases, rather than allowing it to be absorbed into an opaque aggregate engagement metric that makes the tradeoff invisible to outside observers.

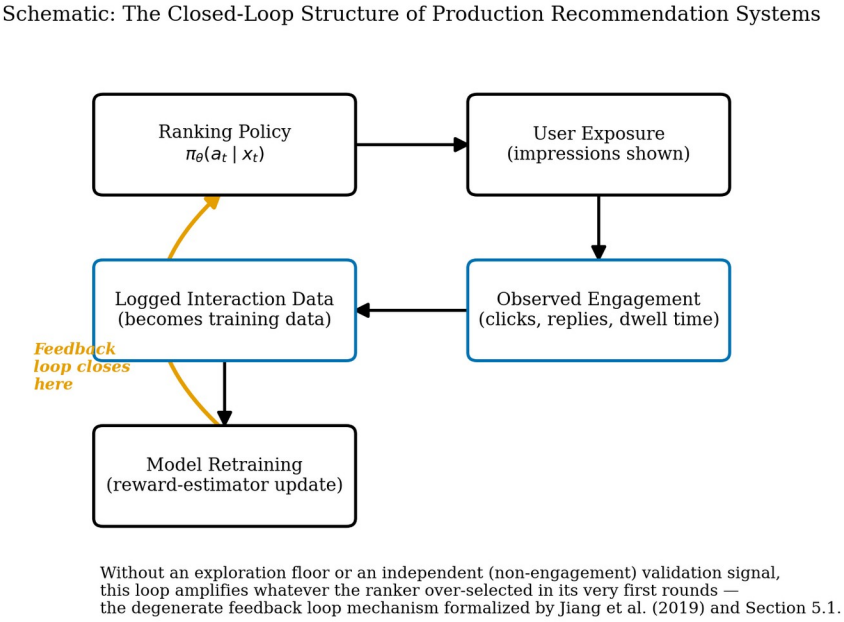


Figure 7. Schematic of the closed-loop structure linking ranking policy, user exposure, observed engagement, logged data, and model retraining. The orange arrow marks where the loop closes: retraining on logged data that was itself generated by the current policy's exposure decisions—the structure formalized as a degenerate feedback loop by Jiang et al. (2019). Without an exploration floor or a validation signal independent of engagement (Sections 2.2 and 5.2–5.3), this loop has no internal mechanism to correct an early skew, regardless of how sophisticated the underlying prediction models are.

5.6 Robustness Across Random Seeds

A single stochastic simulation run risks mistaking sampling noise for a genuine algorithmic effect, so every simulation in this report that compares two algorithms (Figures 3, 5, and 6) was re-run across multiple independent random seeds—eight seeds for the Pareto frontier in Figure 3, six each for Figures 5 and 6—with the shaded regions in each figure showing ±1 standard deviation across seeds rather than a single trajectory. In every case, the qualitative conclusion holds well outside the noise band: in Figure 3, the maximum cross-seed standard deviation in mean

engagement at any point on the frontier is 0.007, two orders of magnitude smaller than the 0.27 total range of the frontier itself. In Figures 5 and 6, the UCB1 and quality-aware Thompson Sampling bands separate cleanly—UCB1's confidence band never overlaps the pool-mean extremity line past round 1,000, and the two engagement-regret bands in Figure 6 stop overlapping by approximately round 2,000—indicating that the divergence between the two algorithms documented throughout Section 5 is a robust property of their designs under this content-pool model, not an artifact of any single stochastic trajectory. We report this explicitly, following standard practice in empirical machine learning, rather than presenting single-run figures as if they were noise-free.

6. A Reproducible Audit Checklist

The preceding sections give a regulator or independent auditor the conceptual vocabulary—vector rewards, scalarization weights, extremity-concentration bounds, exploration floors—needed to ask precise questions about a production ranking system. This section converts that vocabulary into a structured checklist organized into five audit domains, and gives two worked examples of what applying it in practice looks like: a calibration audit (Section 6.1) and an illustrative cross-platform scorecard (Section 6.2). Each checklist item is phrased so that it can, in principle, be answered from disclosed documentation and logged interaction data, without requiring access to a platform's proprietary model weights or training infrastructure; where an item does require log-level data access, this is noted explicitly.

6.1 Worked Example: Auditing Estimator Calibration (Domain D1)

Domain D of the checklist below (Section 6.3) requires that the engagement predictor's calibration be reported by subgroup, not only in aggregate, because an aggregate calibration statistic can mask severe miscalibration on a minority subgroup whose small sample size contributes little to the overall average—the same statistical mechanism our companion paper's Section 2.3 illustrated for facial recognition subgroup error rates. To illustrate what this audit item looks like in practice, we simulated an engagement predictor trained on data disproportionately drawn from a majority-language content subgroup (80% of impressions) and computed subgroup-disaggregated reliability diagrams and expected calibration error (ECE) for both groups, using the standard binned ECE estimator: partition predictions into ten equal-width bins, and take the count-weighted average of the absolute gap between mean predicted probability and mean observed outcome within each bin.

```
def expected_calibration_error(pred, outcome, n_bins=10):
    bins = np.linspace(0, 1, n_bins + 1)
    idx = np.clip(np.digitize(pred, bins) - 1, 0, n_bins - 1)
```

```
    total, weighted_gap = len(pred), 0.0
    for b in range(n_bins):
        mask = idx == b
        if mask.sum() == 0:
            continue
        bin_pred = pred[mask].mean()
        bin_obs  = outcome[mask].mean()
        weighted_gap += mask.sum() * abs(bin_pred - bin_obs)
    return weighted_gap / total

# Applied separately to each subgroup's predictions and outcomes:
# ece_majority = expected_calibration_error(pred[majority_mask], outcome[majority_mask])
# ece_minority = expected_calibration_error(pred[minority_mask], outcome[minority_mask])
```

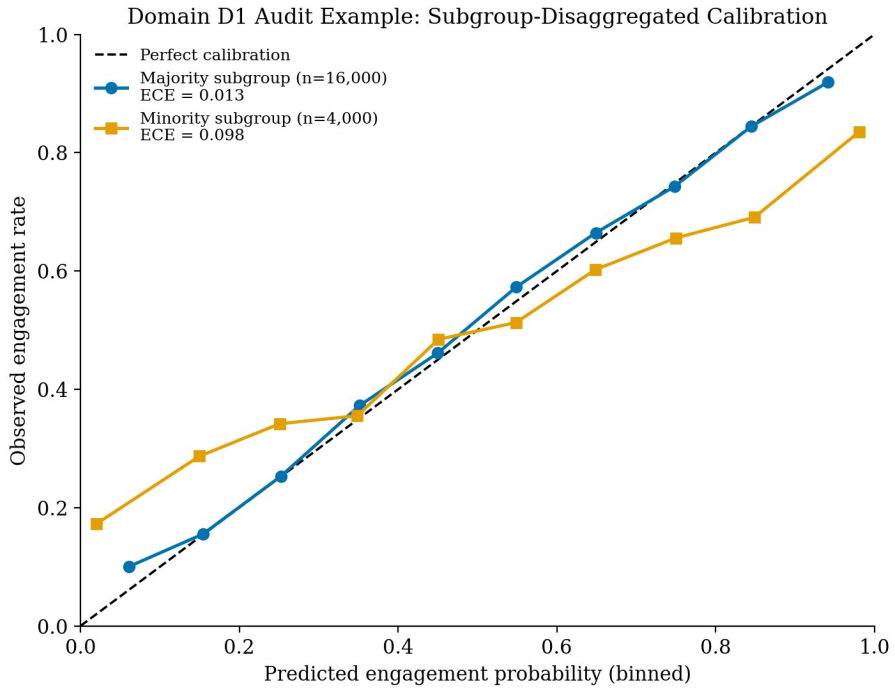


Figure 8. Simulated reliability diagram for a subgroup-disaggregated calibration audit (Domain D1). The majority subgroup (blue, n=16,000, 80% of the simulated population) tracks the perfect-calibration diagonal closely (ECE = 0.013). The minority subgroup (red, n=4,000, 20% of the simulated population) shows systematic overconfidence—predictions pushed toward 0 and 1 relative to observed outcomes—with an expected calibration error 7.7 times larger (ECE = 0.098). An aggregate ECE computed across both groups together would be dominated by the majority subgroup and would substantially understate the minority subgroup's miscalibration.

This worked example is synthetic, not a claim about any specific platform's actual calibration; it is included to make Domain D1 concrete and to demonstrate that the audit item is computationally cheap—it requires only predicted probabilities, observed outcomes, and a subgroup label for each logged impression, all of which a platform could disclose in aggregated, privacy-preserving form (e.g., binned reliability statistics rather than row-level data) without revealing proprietary model internals.

6.2 Worked Example: An Illustrative Cross-Platform Scorecard

A completed audit under this checklist produces, for each of the twenty-two items in Section 6.3, one of three states: verified-satisfied, verified-unsatisfied, or unable-to-verify (typically because the relevant disclosure was not made). Figure 9 illustrates, for three entirely hypothetical platforms constructed only to demonstrate the reporting format, how a completed audit's domain-level results might be visualized in a summary report.

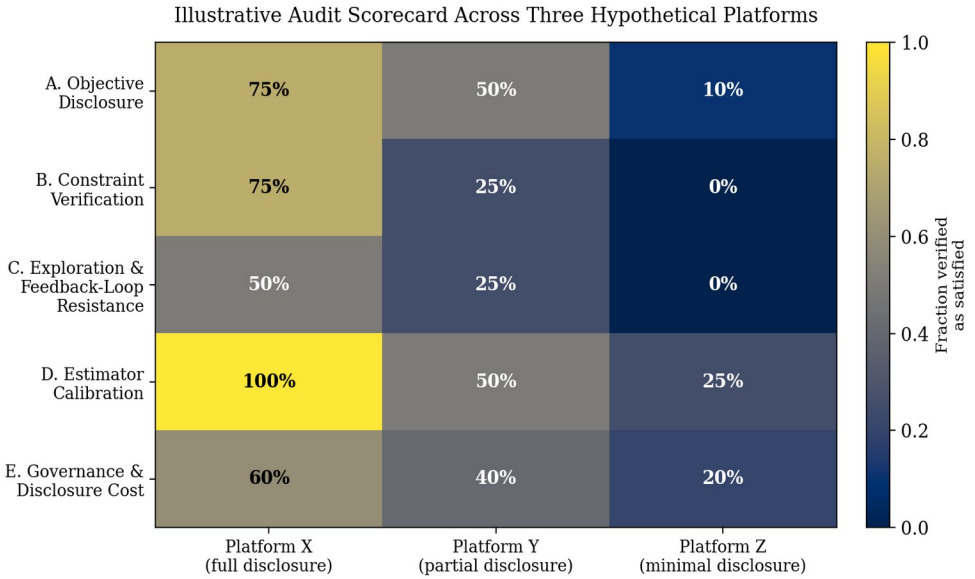


Figure 9. Illustrative audit scorecard format, showing the fraction of checklist items independently verified as satisfied within each of the five audit domains, for three entirely hypothetical platforms ("full," "partial," and "minimal" disclosure) constructed solely to demonstrate the reporting format—the specific percentages carry no evidentiary meaning about any real platform. Domain-level rather than item-level visualization allows a reader to see at a glance where a platform's disclosure is strong (e.g., hypothetical Platform X's Domain D) versus absent (e.g., Platform Z's Domains B and C), while the underlying item-level detail remains available in the full audit report for each domain.

6.3 The Checklist

Domain	#	Audit Item	Data Required
A. Objective Disclosure	A1	Has the platform disclosed the full list of engagement signals (clicks, replies, watch time, etc.) that enter its ranking score, and their relative weights?	Public documentation / regulatory filing
	A2	Are negative signals (blocks, mutes, reports, "not interested") included in the scalarized objective, and at what relative weight versus positive signals?	Public documentation
	A3	Is there an independent quality or accuracy term ($\hat{Q}$) in the ranking objective, and is its estimator architecturally independent of the engagement predictor (Section 2.2)?	Model documentation / architecture disclosure

Domain	#	Audit Item	Data Required
	A4	Is there an explicit outrage or moral-emotional-language term (Ô), and what features does it draw on?	Model documentation
B. Constraint Verification	B1	Does the platform disclose an extremity-concentration bound ε_max (or equivalent), and over what rolling window is it computed?	Public documentation
	B2	Recomputing the realized high-outrage impression rate directly from logged impression data: does it in fact stay at or below the disclosed ε_max?	Impression-level logs (Section 4.2)
	B3	Where the platform uses a scalarized penalty rather than a hard filter, has it disclosed the empirical engagement–extremity tradeoff curve (Section 4.3) rather than only the chosen operating point?	Public documentation / requested disclosure
	B4	Is the quality estimator Q̂ applied retroactively to already-distributed content as corroborating information accumulates, or only to future ranking decisions (Section 2.2)?	Model documentation
C. Exploration & Feedback-Loop Resistance	C1	What exploration strategy does the production ranker use (UCB-family, Thompson Sampling, epsilon-greedy, or a learned exploration policy)?	Model / algorithm documentation
	C2	Is there a disclosed, non-zero forced-exploration floor ρ_min, and what is its value (Section 5.2)?	Model documentation
	C3	Does the platform log and report the rolling mean extremity (or outrage score) of recommended content over time, as in Figure 5, to detect drift?	Time-series impression logs
	C4	Is there evidence, from the recomputed rolling-extremity series, of monotonic drift with no reversion over multi-month windows, consistent with exploration-floor failure (Section 5.1)?	Long-horizon impression logs
D. Estimator Calibration	D1	Is the engagement predictor's calibration (predicted vs. realized engagement rate) reported by subgroup, not only in aggregate (Section 6.1)?	Held-out evaluation data
	D2	Is the quality estimator Q̂ validated against an external ground truth (e.g., independent fact-check agreement rate) rather than only internal consistency?	External validation dataset
	D3	Is the outrage estimator Ô validated against human-annotated samples, and is inter-annotator agreement reported?	Annotation study documentation
	D4	Are all three estimators re-validated on a defined cadence (e.g., quarterly) to detect model drift as language and topics evolve?	Model governance documentation

Domain	#	Audit Item	Data Required
E. Governance & Disclosure Cost	E1	Is the realized engagement cost of any quality or extremity constraint (as in Section 5.4–5.5) disclosed in absolute terms—separately for regret and exposure—not only as a qualitative claim of "improvement"?	Public disclosure / requested filing
	E2	Is there an internal or external escalation path when a rolling-extremity audit (item C3) exceeds the disclosed bound, and what is the documented response time?	Incident-response documentation
	E3	Does the platform offer users a non-profiled or engagement-agnostic feed option consistent with DSA Article 38, and what fraction of users are aware of and use it?	Product documentation / usage statistics
	E4	Are changes to λ_Q, λ_O, ε_max, or ρ_min logged with effective dates, permitting before/after audit comparison?	Change-management logs
	E5	Is the audit methodology itself (this checklist or an equivalent) applied by a party independent of the platform's own ranking team, consistent with DSA Article 37's independent-audit requirement?	Organizational documentation

The five domains move roughly from what a platform says (Domain A: objective disclosure) to what its logs actually show (Domains B–C: constraint and feedback-loop verification) to whether its internal measurement tools can be trusted (Domain D: estimator calibration) to whether the organization is structured to sustain compliance over time (Domain E: governance). An auditor who can only answer Domain A items has confirmed that a platform's stated policy is well-formed; only Domains B through D confirm the policy is actually followed and correctly measured; and Domain E addresses whether compliance is likely to persist as personnel, incentives, and content ecosystems change. A complete audit report should indicate, for each item, whether it was verified from public disclosure, verified from requested but non-public data, or could not be verified—the checklist is a protocol for producing a reproducible confidence level, not a pass/fail certificate. Items E3 and E5 are cross-referenced explicitly to DSA Articles 38 and 37 respectively, since the disclosure obligations those articles already impose on very large online platforms in the European Union substantially overlap with what this checklist would otherwise have to request voluntarily.

6.4 Implementation Overhead of Subgroup-Disaggregated Monitoring at Scale

Domain D's calibration items are computationally cheap to compute once, as Section 6.1 showed, but checklist item D4's requirement of a defined re-validation cadence raises a distinct

question this report has not yet addressed: what does it cost a platform operating at production scale to maintain subgroup-disaggregated expected calibration error (ECE) as a continuously monitored quantity rather than a one-time audit artifact? Three overhead sources are worth distinguishing. First, storage: computing ECE by subgroup requires retaining a subgroup label alongside every logged (prediction, outcome) pair, which multiplies logging volume by the number of tracked subgroup dimensions if labels are not already collected for other purposes (e.g., localization); platforms already collecting language or region metadata for product reasons incur near-zero marginal storage cost, while platforms that would need to newly infer sensitive subgroup labels face a materially harder problem that item D1 does not by itself resolve. Second, compute: the binned ECE estimator in Section 6.1's code sample is O(n) per subgroup per evaluation window and trivially parallelizable, so compute cost scales linearly with the number of monitored subgroups and is unlikely to be the binding constraint for any platform already running the engagement predictor itself at the same data scale. Third, and most significant in practice, statistical power: a subgroup comprising a small fraction of traffic, as in Section 6.1's 20%-share minority example, requires a correspondingly longer accumulation window before its per-bin sample sizes are large enough for a stable ECE estimate—for the ten-bin estimator used here, a subgroup would need on the order of several thousand impressions per bin for the reported ECE to be estimated within a reasonably tight margin, meaning real-time monitoring of small subgroups is inherently a lagging rather than a leading indicator. A pragmatic implementation would compute high-traffic subgroup calibration on a rolling daily or weekly cadence, as item D4 requires, while accepting that the smallest subgroups may only support monthly or quarterly re-estimation—a limitation that should itself be disclosed as part of a complete Domain D response, rather than left implicit in a single reported cadence that in fact varies by subgroup size.

7. Limits of a Mathematical Audit Framework

7.1 Internal Consistency Is Not External Adequacy

Consistent with the discussion in our companion paper, it is important to be explicit about what this framework does not do. Verifying that a platform's logged behavior is consistent with its disclosed ε_{max}, λ_Q, and λ_O (Domain B) establishes internal consistency, not external adequacy: a platform could satisfy every item on this checklist while having chosen an ε_{max} that a reasonable observer would consider far too permissive, or a quality estimator that is technically independent of the engagement predictor but trained on data that encodes its own biases. The checklist converts a normative disagreement about acceptable tradeoffs into a precise, quantified disagreement—exactly where on the Figure 3 frontier a platform should operate, or exactly how

much of the Figure 6 regret–exposure tradeoff it should accept—but it cannot resolve that disagreement, because the correct operating point depends on value judgments about acceptable engagement cost that are not themselves mathematical questions.

7.2 Estimator Generalization and Subgroup Coverage

Every estimator in Section 2.2 is itself a model, trained on finite data, subject to the same statistical limitations discussed throughout the broader algorithmic-fairness literature this framework draws on (Mehrabi et al., 2021): a quality estimator trained predominantly on English-language, Western news sources will generalize poorly to other linguistic and cultural contexts, and an outrage estimator trained on one platform's data may transfer poorly to another's user base or content format—the same failure mode illustrated synthetically in Section 6.1's calibration audit. Validating an estimator is necessary, not sufficient, for trusting it in deployment, and a full audit would need to extend Domain D with subgroup-disaggregated calibration checks across many more subgroup definitions than the single illustrative split used in Section 6.1.

7.3 Adversarial Co-Adaptation: Threshold Gaming by Content Creators

The fair-ranking and constrained-bandit literature reviewed in Section 3 was developed primarily for settings with a clear, comparatively static notion of item merit—candidate qualifications in a hiring context, document relevance in a search context—where the item being ranked has no strategic incentive to reshape itself in response to the ranking rule. The setting this report addresses is different in a way that matters: $\hat{O}(c)$ is not a fixed physical property of content but a function of features (moral-emotional language density, out-group references, predicted anger reaction) that a content producer can observe indirectly, through the ranking outcomes and engagement their posts receive, and adjust. This turns the audit problem into a two-player dynamic rather than a static measurement problem. A content producer motivated to maximize reach effectively runs their own optimization against $\hat{O}$: holding the emotionally engaging core of a post fixed, they can iteratively reduce its measured moral-emotional language density, remove explicit out-group references, or otherwise edit surface features until the post scores just under τ, while leaving the underlying arousal or divisiveness that actually drove engagement largely intact. We term this an adversarial co-adaptation loop: the platform updates $\hat{O}$ (or the threshold τ) in response to observed content, and content producers update their content in response to observed $\hat{O}$ scores or rejection patterns, each side's optimization changing the landscape the other is optimizing against.

This dynamic has a direct precedent in the adversarial-example and spam-filtering literature, where classifiers trained on a static snapshot of adversary behavior degrade as the adversary adapts, and the standard responses from that literature apply here with some modification. First, a

static, publicly inferable threshold τ is a liability: if τ is fixed and content producers can estimate it empirically by probing the system, the checklist's Domain B verification (recomputing realized high-outrage rates from logs) will show apparent compliance even as the true, human-perceived outrage of distributed content stays high, because Ô has been gamed rather than genuinely reduced. A more robust design randomizes τ within a disclosed band, or maintains an ensemble of outrage estimators with periodically rotated feature sets, so that no single, stable target exists for producers to optimize against—at the cost of making the constraint itself slightly less precisely interpretable for audit purposes, a tradeoff an auditor should explicitly ask about under an extended version of checklist item C1. Second, periodic human-annotated re-validation of Ô (checklist item D3) is doing double duty in this setting: beyond checking ordinary calibration drift, it is the mechanism that would detect a widening gap between Ô's automated score and genuine human-perceived outrage, which is precisely the signature adversarial co-adaptation would produce. A checklist that only re-validates estimators against automated benchmarks, rather than fresh human judgment, is structurally blind to this failure mode. We do not attempt to formalize the equilibrium of this co-adaptation game here—doing so rigorously would require a model of content-producer incentives and adaptation costs beyond this report's scope—but flag it as the single most important respect in which the static estimators specified in Section 2.2 understate the operational difficulty of maintaining a durable extremity-concentration bound in an adversarial production environment.

7.4 The Exogenous Choice to Optimize for Engagement

Finally, this framework, like our companion paper, treats the choice to optimize for engagement at all as exogenous; explaining why that choice was made, and whether it should be regulated directly rather than merely constrained at the margin, remains a question about platform business models and incentives that a bandit specification and an audit checklist can inform but cannot settle on their own.

8. Conclusion

This report has converted the diagnostic argument of our companion paper into an engineering and auditing specification: a formal multi-objective bandit that makes quality and outrage first-class terms in the ranking objective rather than afterthoughts, situated explicitly within the established fair-ranking and constrained-bandit literature (Sections 2–3); a Pareto-optimal ranking filter that bounds extremity concentration by construction and whose engagement cost can be characterized empirically as a frontier rather than assumed away (Section 4); an analysis showing that the choice of exploration strategy is not incidental to the amplification problem but actively compounds or resists it, together with a quality-aware Thompson Sampling

variant whose cost we decomposed into separate, independently interpretable engagement-regret and outrage-exposure curves (Section 5); and a twenty-two-item, five-domain audit checklist, illustrated with two worked examples—a calibration audit and a cross-platform scorecard—designed to be answerable from disclosed documentation and logged data by a party independent of the platform's own ranking team (Section 6). Together with the honest accounting of this framework's limits in Section 7, we intend this as a template for the kind of quantitative, reproducible specification that would need to underlie any regulatory or platform-governance response to algorithmic amplification that goes beyond content moderation to address the objective function itself—the level at which, as our companion paper argued, and as the independent, massive-scale evidence in Huszár et al. (2021) corroborates, the problem actually originates.

References

Brady, W. J., Wills, J. A., Jost, J. T., Tucker, J. A., & Van Bavel, J. J. (2017). Emotion shapes the diffusion of moralized content in social networks. Proceedings of the National Academy of Sciences, 114(28), 7313–7318.

Celis, L. E., Kapoor, S., Salehi, F., & Vishnoi, N. K. (2019). Controlling polarization in personalization: An algorithmic framework. In Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT* '19), 160–169.

European Commission. (2022). Regulation (EU) 2022/2065 of the European Parliament and of the Council of 19 October 2022 on a Single Market For Digital Services (Digital Services Act), Articles 27, 37, and 38.

Huszár, F., Ktena, S. I., O'Brien, C., Belli, L., Schlaikjer, A., & Hardt, M. (2021). Algorithmic amplification of politics on Twitter. Proceedings of the National Academy of Sciences, 119(1), e2025334119.

Jiang, R., Chiappa, S., Lattimore, T., György, A., & Kohli, P. (2019). Degenerate feedback loops in recommender systems. In Proceedings of the 2019 AAAI/ACM Conference on AI, Ethics, and Society, 383–390.

Joseph, M., Kearns, M., Morgenstern, J. H., & Roth, A. (2016). Fairness in learning: Classic and contextual bandits. Advances in Neural Information Processing Systems, 29.

Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K., & Galstyan, A. (2021). A survey on bias and fairness in machine learning. ACM Computing Surveys, 54(6), 1–35.

Merrill, J. B., & Oremus, W. (2021, October 26). Five points for anger, one for a ‘like’: How Facebook’s formula fostered rage and misinformation. The Washington Post.

Russo, D. J., Van Roy, B., Kazerouni, A., Osband, I., & Wen, Z. (2018). A tutorial on Thompson sampling. Foundations and Trends in Machine Learning, 11(1), 1–96.

Satkynov, I., Abdyrasulova, B., & Krutchenko, I. (2026). Engagement as objective: How the mathematics of recommendation algorithms systematically favor extremity. [Companion paper].

Singh, A., & Joachims, T. (2018). Fairness of exposure in rankings. In Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD '18), 2219–2228.

Singh, A., Kempe, D., & Joachims, T. (2021). Fairness in ranking under uncertainty. Advances in Neural Information Processing Systems, 34, 11896–11908.

Sutton, R. S., & Barto, A. G. (2018). Reinforcement learning: An introduction (2nd ed.). MIT Press.